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Taxonomy Based Question Generation Using Prompt Engineering for Student Assessment

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***Abstract*—The recent wide popularity of ChatGPT has resulted in immense research interest in Large Language Models in many application areas such as education, business, healthcare, and tourism. Its popularity is due to its human-like conversation and comprehension capability. In this paper, we focus on the usage of generative AI in the education system to address the challenges of generating questions to address the different levels of learning taxonomy and help educators design questions based on the different categories of students. In this work, we are trying to build an AI system that is capable of generating questions based on Bloom's Taxonomy using prompting techniques. The system is also capable of generating personalised quizzes for students who face difficulty answering learning taxonomy-based questions. We have used the LLM API and four types of prompting techniques to generate questions in our personalised evaluation system. The paper presents the system architecture of the personalised evaluation system. The paper also presents a comparison of various prompting techniques, and the result obtained.**

***Index Terms*—LLM, Generative AI, Prompt Engineering, Education, Learning Taxonomy, Evaluation System.**

I. Introduction

There are different kinds of learners in the student community and a single type of teaching-learning method is not suitable for all type of learners. Researchers have been focusing on producing learner-centric approaches which will foster student learning. The researchers experimented with flipped classrooms approach [1], recorded video lectures [2], e-learning [3], online learning [4] etc. to foster students learning. Many Massive Open Online Courses (MOOCs) and large-scale blended interactive classrooms [4] were developed and introduced to support a larger student community. However, several such MOOCs suffered from high dropout rates and a lack of engagement [5]. Researchers are exploring new ways to engage students in the current distractive environment created by the social media and instant messaging era. Hence it is important to design a highly engaging and personalized learning environment for students. Currently, the role of AI is expanding to the education area to address the challenges faced by the students and educators in the traditional teaching-learning methodology. AI has great potential to address the need for personalized learning such as personalized content generation, and personalized quizzes with the help of existing learning taxonomies such as Blooms [6] and Marzanos (taxonomy for higher order thinking) [7].

Currently, generative AI and Large Language Models (LLMs) are capable of producing generative content for several applications. One of the main promises of generative AI in education is its ability to provide personalized learning experiences according to the preferences and abilities of the user [8]. This helps to eliminate the traditional approach of one-size-fits-all instructional methods. Adaptive algorithms can provide personalized educational materials to enhance comprehension and foster engagement.

In the education sector, assessments are meant to understand the gaps in student learning and they help educators to adjust their pedagogy to suit the learner's strengths and weaknesses. LLM and the latest AI-generated questions by using taxonomy such as Bloom's level may help both students and teachers in achieving high-quality assessments. Such high-quality assessment helps students to relate their learning to the real world and also to engage with the content [9]. Traditional approaches such as single-question papers based on topics covered in class may not address all these student achievements. Educators have to spend plenty of time

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developing questions that promote learning and Automated Educational Question Generation (AEQG) systems can minimize the cognitive load and effort of educators [9]. Another study [10] used unsupervised machine learning to support and enhance personalized learning, adaptive testing, and question generation by accurately mapping difficulty levels in question banks. Similarly, automatic assessment item bank calibration has been proposed in [11], to facilitate precise learning gap identification, thereby reducing the manual effort required from educators. The major challenge in the current education sector is the need to adaptively generate questions that are personalized for individual learners.

In this work, we are focusing on high school education to provide AI-based solutions to address personalized quiz generation requirements. This paper presents various scenarios in which an AI-enabled personalized evaluation system helps both students and educators. The paper also details the system architecture and how prompt engineering is being utilized for generating personalized questions. To generate a questionnaire, we employ the Google Generative AI API (google.generativeai). We have employed prompt engineering techniques to generate questions for the content selected for the quiz. The prompt engineering approach here ensures that the model understands the format and type of questions required. The tags "*Short*" and "*Long*" have been used to help the model differentiate between short and long answer questions.

Section II discusses the literature review. Section III discusses the requirement of personalized quiz generation and also details the system architecture. Section IV discusses the implementation. Section V discusses results and Section VI presents the conclusion of the work.

II. LITERATURE REVIEW

A large language model is a deep learning model that is trained on a large dataset and can generate text that is similar to human language [12]. This can be utilized for improving the education sector. Many existing studies focus on the use of AI for question generation in the education area. Many of the existing works utilized GPT-3 and other generative AI techniques for question generation. Research-based on automatic question generation is classified into two groups. One is to automate the educational assessment and the other one is to support the development of an interactive question-and-answer system [13].

In one of the works, authors tried to investigate whether given an arbitrary educational text as input, *Is it possible to automatically generate quizzes to use as formative or summative feedback in an educational setting?* [14]. The researchers proposed EduQuiz which is a GPT-3-enabled end-to-end quiz generator, that generates quizzes on new input texts. Different techniques, such as rule-based, template-based, and neural networks-based approaches, were used for automatic question generation. These methods extract a single sentence or multiple sentences or a context from the input paragraph, which is then converted into a question [5].

Research also shows that prompt engineering can play a major role in automatic quiz generation. The authors [15] discussed Electronic Quiz Generation using two prompting techniques with GPT-3.5. They initially use a simple prompting strategy that uses a generic strategy to generate questions. They have also used a controlled prompting strategy, which uses taxonomy to generate questions with different learning goals in mind. Individual aspects that determines the quality of a quiz such as coverage, structure, redundancy, and usefulness were measured using a set of defined metrics. Relevancy, fluency, and answerability metrics were also outlined to evaluate the quality of individual questions in a quiz [15].

The paper [16] introduces a multi-dimensional evaluation benchmark that helps to improve questions generated by various Question Generation Models. The design helps to evaluate questions through seven dimensions i.e. fluency, clarity, conciseness, relevance, consistency, answerability, and answer consistency. The dataset contains several questions generated by different question generation systems, which help to improve the evaluation. Pearson correlation was used to show the correlation between automatic metrics and human evaluation.

Another work [17] proposed a Deep Question Generation (DQG) problem, which can generate complex questions that require reasoning to answer. The proposed model consists of four parts: (i) an encoder that encodes the input document, (ii) an encoder that encodes the semantic graph through the Attention-based Gated Graph Neural Network, (iii) a content selector to select relevant questions, and (iv) a question decoder to generate questions. Experiments conducted using HotpotQA dataset demonstrate that the introduction of a semantic graph significantly reduced the semantic errors, leading to better-quality questions. Another work on AI to create and evaluate MCQs in introductory chemistry and biology is discussed in [18]. Questions are generated using OpenAI's GPT-3.5-turbo model. The prompt was used to generate five MCQs with one correct answer and three distractors for each level of Bloom's taxonomy.

A use case of AI in education can be found in the paper [19] which focuses on generating reading comprehension quizzes to help teachers. They have used Natural Language Generation (NLG) metrics to evaluate the questions generated by the models. They have also invited experienced teachers to design quizzes composed of 8-12 questions using the reading material and 7 automated QGen models. Teachers review and select questions generated by the QGen models that are included in the quiz.

LLMs can be utilized in different ways in education, by creating personalized content, helping with assessments, and providing feedback [9]. LLMs are capable of generating relevant educational questions according to the different cognitive levels. The different LLM prompting techniques such as Role-Playing (RP) and (ii) Chain-of-Thought (CoT) prompting allow LLM to generate content based on role and reasoning [20]. The research [9] uses five different LLMs to examine high-quality and diverse AI-generated questions across

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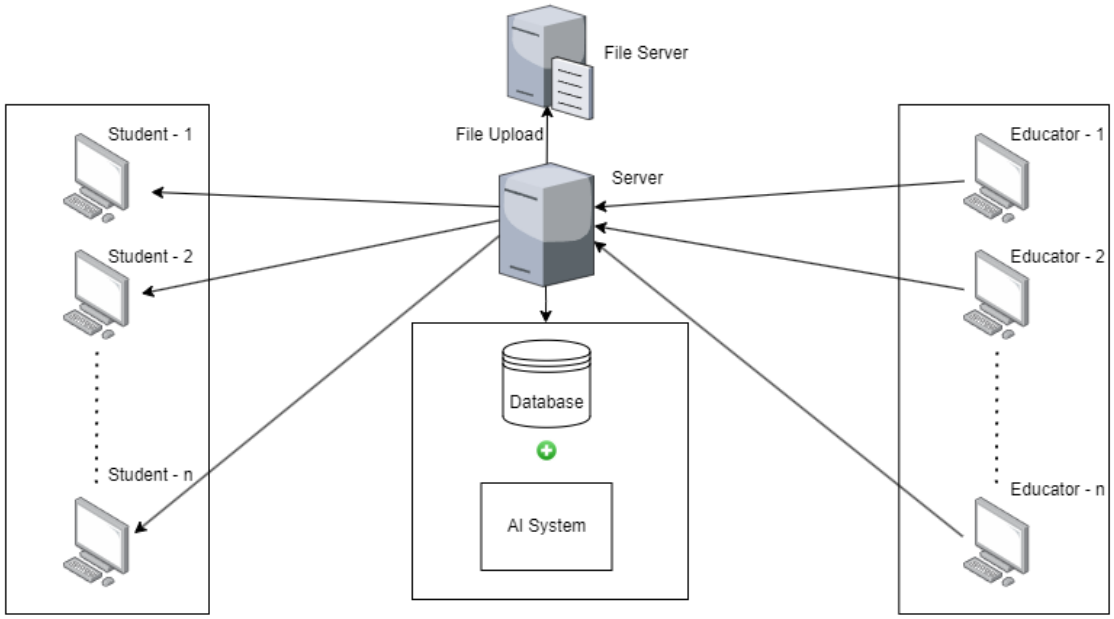

Fig. 1. Proposed Personalized Evaluation System Architecture

various cognitive levels, defined by Bloom's taxonomy. The experiments utilized prompting techniques like zero-shot, few-shot, and Chain of Thought(CoT), with varying complexity for automated educational question generation. Linguistic and pedagogical quality was evaluated using the nine-item rubric.

III. Personalized Quiz Generation : Requirements and System Architecture

This section presents the learner's requirements, learner's classification, and system architecture.

A. Learner Requirements

We present different scenarios in which a student might want support which can propel their learning rate and knowledge acquisition. Students learn best while they are attentive in class and by revising the concepts covered in class. For this, the students should be provided with revision notes, core concepts reviser option, catching up on missed classes, self-tests, and corrections to ensure the concepts are learned properly. Every topic might not be interesting to students at first and hence engaging students in subjects and topics is also crucial for the overall enhanced learning outcome and is explained below:

1) Revision Notes: Quick revision notes that can be generated for the required chapters so that the student can save time on revision. Students can find it hard to revise already covered chapters and may not be able to recollect points as effectively as they used to. In this scenario our app can be used to generate revision notes with only the relevant information and each topic discussed in the chapter should be summarized in points so that the student can get the whole idea of the chapter with a glance over this notebook.

2) Core concepts refresher: Students may find it difficult to recollect core concepts once the teacher moves to the more advanced topics. A short one-shot learning note that can keep track of the necessary core concepts of the materials currently updated by the teacher can help the student quickly recollect the core concepts and attend the class more effectively.

3) Catching up on Missed Classes: The generated workbook helps to cover the materials that the student missed. The app provides a detailed explanation of the topics and generates practice problems to help students catch up on missed parts.

4) Test Correction and Feedback: AI can assist in the correction process by providing detailed feedback on where mistakes were made and suggesting necessary exercises to correct them. The AI system analyzes test responses, identifies specific errors, and generates personalized feedback for each student.

5) Engagement with low-interest topics: AI can make these topics more engaging by presenting topics in various innovative ways. This approach can help students develop a greater interest in subjects they initially found unappealing by making the learning process more dynamic and enjoyable.

6) Unanswered questions could have been answered if phrased differently: Sometimes, each student understands a question differently, and unanswered questions often stem from unclear or complex phrasing. If these questions had been rephrased differently, the student might have understood them better and tried to answer them.

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7) Personalized explanation based on student's interest: If a student struggles to understand a topic, then the system explains the concept using an analogy related to the student's interests, making it easier to understand. This personalized approach can enhance learning and engagement.

B. Learner-Category Requirement

There are two types of user categories for the system, including students and educators. There can be multiple educators for the various topics/subjects. Each educator has a login and he can connect to the system. Educators can upload their study material into the system and can use the evaluation system to generate quizzes based on Bloom's taxonomy. The second category of users is students; they can log in using their devices and will be able to see the quiz intended for them. Once students select the quiz in the system, they will be able to see the questions generated by the LLM API and approved by the educator. The student will have the flexibility to engage different types of questions in case he/she is finding it difficult to understand the question. The predetermined question will be rephrased with different examples and given to the student for better understanding with a minor penalty.

C. System Architecture : Learner-Centric Evaluation System

We have proposed a client-server architecture for our system since it offers centralized management and scalability. This ensures that all quizzes, educational materials, and user data are efficiently handled on the server and can facilitate the organization of resources. The system should be scalable to accommodate an increasing number of users. Fig.1 represents a client-server architecture of our proposed system. The users can access the system using their client login. The system is composed of an AI system that generates quizzes and a database to store the quizzes. The database also stores the user details i.e. the student and the educator details, the generated quizzes, the students responses, and the evaluation details. The server handles the user queries and provides the response.

There is a file server that handles the study materials, and educators can upload the study materials to this server and they will be stored for future use. The AI system will use the materials in the file server based on the educator's requirement and will present the questions generated by the LLM-enabled AI system. The educator can approve the question and can use it for the quiz. Once the questions are finalized, the educator can schedule the quiz, and the intended student group can view it from their client login.

IV. Implementation

As part of the implementation, we have used the first chapter from the NCERT history book "Our Pasts III" of class VIII as input text. We have generated 15 questions using four prompting methods with the help of LLM. The limited availability of educational data and high training costs can make training a language model from scratch or fine-tuning existing models an expensive process [9]. To produce optimal results from the Large Language Model (LLM) and address the challenges of high training costs, we use prompt engineering. Specifically, we used zero-shot and Chain-of-Thought (CoT) prompting techniques with the LLM API to generate questions, using this we can enhance the performance of generative AI models, ensuring that the generated questions meet our desired criteria. The whole process of the implementation step is shown in Fig. 2. We also compare the quality of questions generated by the different prompting techniques.

A. Dataset

Our implementation considers the first chapter of the NCERT history book "Our Pasts III" of class VIII. The data set was uploaded as a file and the system extracted text from it and used that as the content for the prompt. We have generated 15 questions using four prompting methods using the Gemini API.

The overall implementation is done as two modules: the text extraction and the question generation module.

B. Text extraction from materials

Our implementation utilizes the PyPDF2 library to extract text from the PDF file. It is an open-source PDF library, that can split, merge, crop, and transform the pages of PDF files. This library is used to read the PDF file, iterate through each page, and collect the extracted text, both the PDF file and the extracted text are saved inside Amazon's Simple Storage System (S3) bucket database for easy storage and retrieval. S3 is capable of storing objects of variable sizes. The client application can read and update these S3 objects remotely either through a SOAP or REST-based interface [21].

C. Generating questions from the extracted text

For implementation, we have used Gemini API (LLM) to generate questions from the extracted text. Google Gemini API enable users to integrate Gemini model functionalities into their applications [22]. Gemini is based on Bidirectional Encoder Representations from Transformers (BERT), which uses a transformer architecture, and processes words in relation to all other words in a sentence instead of one by one in order. It helps to get a deeper understanding of the context [23] as it is pre-trained on a large English corpus [24]. BERT is pre-conditioned with two goals Masked Language Modelling (MLM) and Next Sequence Prediction (NSP). Gemini, the latest multimodal artificial intelligence (AI) tool launched by Google's DeepMind on December 06, 2023, with Visual Language Model (VLM) technology. Gemini can understand and work text, images, audio, PDFs, and videos [25].

To complete the experimentation, we have created an environment setting file called "env" which is used to save all the secret API_KEYs to access LLM API. Then set a variable named API_KEYs and set its value to our API key. We have also imported $google.generativeai$ as genai and used $load_dotenv$ function from the dotenv library to configure the gemini api key value to the one we have set in the $.env$ file. We have then set the generative model used as "gemini-1.5-flash". Various types of prompting techniques were used for our study. The details are given below.

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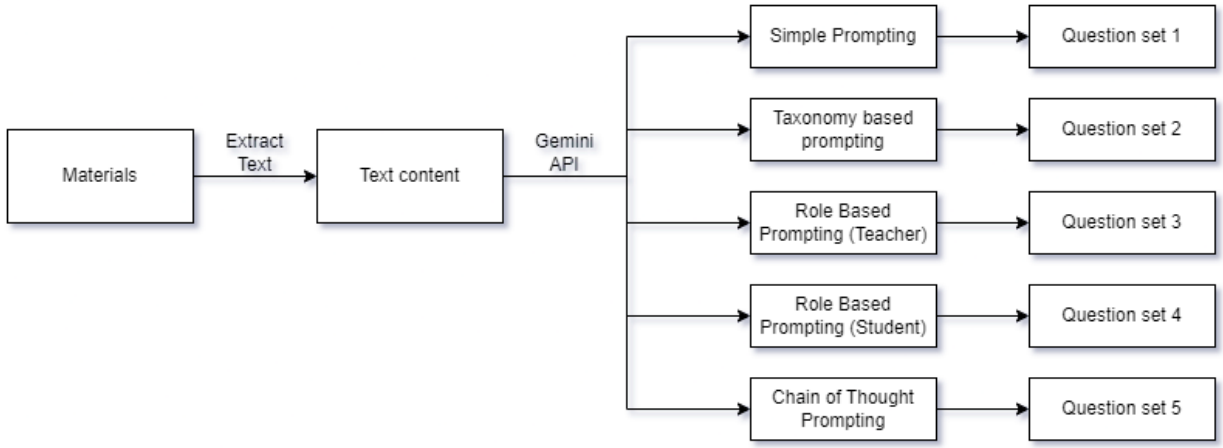

Fig. 2. Question generation process using Gemini API and prompting

D. Prompt Design

Four types of prompting techniques are used to conduct the experiment. Each prompt result is been compared and tested with the educators and took their feedback for system building. The different prompting techniques used and the strategies applied are given below:

*1) **Normal prompting**:* To generate questions, initially, we provided a straightforward instruction based on the input text. The prompt used is "Create 15 questions based on the following text". This simple prompt shown in Fig. 3 has generated 15 questions from the provided input text.

```
"Normal prompt:\n"
"We have extracted the text from a textbook.\n"
"Generate 15 questions based on the following text:\n"
f"(file_content)"
```

Fig. 3. Normal prompt using the LLM

```
"Prompt based on Bloom's taxonomy:\n"
"We have extracted the text from a textbook.\n"
"Generate 15 questions based on the following text,
explicitly aligning each question to a specific level
of Bloom's taxonomy: \n"
"Knowledge: Recall of facts, terms, basic concepts\n"
"Comprehension: Understanding the meaning of
information\n"
"Application: Using learned knowledge in new
situations\n"
"Analysis: Breaking down information into parts to
explore understandings\n"
"Synthesis: Combining information to form a new whole
\n"
"Evaluation: Judging the value of information\n"
"Also specify the level of Bloom's taxonomy for each
question.\n\n"
f"(file_content)"
```

Fig. 4. Instruction prompt using LLM

*2) **Instruction Based Prompting**:* In the Instruction based prompting approach, the prompt given to LLM API has a set of instructions that provides a detailed breakdown of the structure of the content that we expect from the generated questions. The prompt is shown in Fig. 4, which also contains a short description for each level of Bloom's taxonomy and instructs the model to generate 15 questions that align explicitly with the various levels of Bloom's taxonomy.

*3) **Role-Based Prompting**:* To generate questions using role-based prompting, a prompt is assigned with a specific role, which influences the type and tone of questions generated by the LLM. Since the system is to be majorly used by educators and students, we have experimented with two roles: an educator and a student. For educator role-based prompts, the model generated questions that reflected the style and tone of educators. The prompt is shown in Fig. 5, which instructs the LMM to generate questions with varying difficulty levels that can assess the student's understanding of the provided text.

```
"Role-based prompting - teacher:\n"
"We have extracted the text from a textbook.\n"
"You are a knowledgeable teacher. Create 15 questions
of varying difficulty levels to assess students'
understanding of the following text:\n"
f"(file_content)"
```

Fig. 5. Utilizing the educator role for Role Prompting

```
"Role-based prompting - student:\n"
"We have extracted the text from a textbook.\n"
"You are a curious student. Based on the following
text, generate 15 questions you would like to ask:\n"
f"(file_content)"
```

Fig. 6. Utilizing the student role for Role Prompting

The second experiment was student-centered, and the prompt was given from the student role as shown in Fig. 6. The prompt was instructed to generate 15 questions, which mimic the questions in the mind of a student who is seeking a deeper understanding of the concept. The generated questions

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were mostly exploratory in nature and focused on areas that can cause either interest or confusion.

*4) **Chain of Thought (CoT):*** Chain-of-Thought is a prompting technique, which performs complex reasoning by generating intermediate reasoning steps [26]. Manual step-by-step instructions were added for each of the above prompts to generate more relevant questions. As seen in Fig. 7, the prompt guides the LLM through step-by-step instructions that help to identify key points, formulate question words, and construct questions. This method filters out the necessary questions much more efficiently to generate questions that are much more refined and relevant.

```
"COT-Bloom's taxonomy prompt:\n"
"We have extracted the text from a textbook.\n"
"Generate 15 questions based on the following text,
aligning each question to a specific level of Bloom's
taxonomy. Use the following steps:\n"
"1. Identify key points in the text.\n"
"2. Determine the appropriate Bloom's taxonomy level
for each key point.\n"
"3. Formulate a question based on the key point and
Bloom's level.\n"
"Knowledge: Recall of facts, terms, basic concepts\n"
"Comprehension: Understanding the meaning of
information\n"
"Application: Using learned knowledge in new
situations\n"
"Analysis: Breaking down information into parts to
explore understandings\n"
"Synthesis: Combining information to form a now whole
\n"
"Evaluation: Judging the value of information\n"
f"(file_content)"
```

Fig. 7. COT prompt to generate questions based on blooms taxonomy

The results of each of the prompting techniques is discussed in the result section.

V. Results

In this research, we compared the performances of four different prompting strategies using the Gemini model to generate questions. It is observed that each strategy influenced the nature, structure, and cognitive depth of the questions generated. The details highlighting the effectiveness and unique characteristics of the generated questions are presented below:

1) Normal Prompting: The questions generated using this prompt were well-structured and evenly distributed across the selected chapter. However, the generated questions were direct and lacked depth in cognitive engagement. The questions also exhibited redundancy, with overlapping questions repeating similar aspects of the selected chapter. This clearly shows the limitation of this prompting approach for question generation.

2) Bloom's Taxonomy-Based Prompting: In this prompting method, all six levels of Bloom's Taxonomy were covered by the generated questions. This showcases a comprehensive approach to assess different cognitive skills of the learner. Questions, for each taxonomy level had a distinctive structure according to the specific cognitive domain. This highlights the LLMs capability to generate well-structured and pedagogically sound questions based on the different levels of Bloom's taxonomy.

3) Teacher Role-Based Prompting: The generated questions using this prompt were well balanced with different types of questions, including true/false, identification, summarization, and analysis. LLM identified three main assessment levels automatically without providing any specifications and generated questions that followed a structured format, resembling real-world classroom assessments.

4) Curious Learner Role-Based Prompting: The generated questions using this prompt encourages deeper thinking and engagement with the chapter by being exploratory in nature. Additionally, the questions closely resemble those that a student might ask when seeking clarification, demonstrating the model's ability to generate inquiries that align with a learner's perspective.

5) Chain-of-Thought (CoT) Prompting: The prompt demonstrated an improved ability of LLM to identify key points in the text before generating questions. This led to more structured and contextually relevant outputs, aligned with the provided content. Moreover, the step-by-step instruction of CoT minimized redundancy and improved clarity of the generated questions.

TABLE I
Comparative Analysis of Prompting Strategies Based on Key Factors

Prompting Method	**Cognitive Depth**	**Question Variety**	**Alignment with Provided Text**	**Redundancy**
Normal Prompting	Low	Moderate	Moderate	High
Bloom's Taxonomy Prompting	High	High	High	Low
Teacher Role-Based Prompting	Moderate	High	Moderate	Moderate
Curious Learner Prompting	Moderate	High	Moderate	Moderate
CoT-Based Prompting	High	High	High	Low

Results clearly exhibits the capability of LLM to produce taxonomy-based and contextually relevant questions. The results obtained for the different prompts based on some key factors such as cognitive depth, variety, redundancy, and alignment with the provided content is shown in Table I. Here cognitive depth is high if the generated questions spans across all levels of Blooms taxonomy and moderate if the questions contains more than 3 levles of taxonomy and low if questions are from the first two levels of Blooms taxonomy. We classified question variety high, if multiple types of questions were generated and moderate only descriptive questions. Similarly, if the generated questions are aligned with the content provided for question generation, we classify it as High, and moderate if questions are mostly aligned with the content but have general references outside the provided text. Redundancy is high if

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many questions are repeating, moderate when there are some overlaps, and low if the questions are unique and distributed across the topics.

It is evident that Bloom's Taxonomy-Based along with CoT prompting gave us better questions suited to test all levels of learners compared to other methods. However, the generated questions lacked diversity within the same taxonomy level in Bloom's Taxonomy-Based prompting. Teacher role-based prompting successfully created classroom-style questions suited for assessment, but it occasionally overemphasized factual recall over higher-order thinking. Curious learner prompting resulted in exploratory, open-ended questions resembling real student inquiries, yet some questions were too vague or lacked focus. CoT prompting significantly improved question relevance and minimized redundancy, further emphasizing the importance of structured instructions.

VI. Conclusion

Learner-centric evaluation systems are necessary to create personalized learning environments for students. This paper focused on addressing the need for an AI-enabled personalized learner-centric evaluation platform to foster learning and empower the student community. We have proposed a learner-centered platform for students to attend online tests to understand their learning levels. The system architecture and the question generation using the latest prompt engineering and the steps used to generate questions have been presented. The system allows the students to test themselves on the topics they have learned in their school. The paper also presented the four different types of prompt engineering techniques used to generate the questions. Compared to the performance of these prompting methods, it is found that the Chain-of-Thought-based prompt created the best questions. However, for applying it to different subjects, detailed study of customized prompting strategies are needed, which remains as a future scope of the study.

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References

[1] K. Bijlani, S. Chatterjee, and S. Anand, "Concept maps for learning in a flipped classroom," in *2013 IEEE Fifth International Conference on Technology for Education (t4e 2013)*. IEEE, 2013, pp. 57–60.

[2] S. Anand, S. Chatterjee, and K. Bijlani, "Pedagogy experiments with recorded video lectures," in *2014 IEEE Sixth International Conference on Technology for Education*. IEEE, 2014, pp. 193–194.

[3] N. S. Subramanian, S. Anand, and K. Bijlani, "Enhancing e-learning education with live interactive feedback system," in *Proceedings of the 2014 International Conference on Interdisciplinary Advances in Applied Computing*, 2014, pp. 1–7.

[4] R. A. Kumar, T. S. Soman, S. Narayanan, and K. Bijlani, "Hybrid architectures for large-scale blended classrooms: A case study," in *2018 9th International Conference on Computing, Communication and Networking Technologies (ICCCNT)*. IEEE, 2018, pp. 1–7.

[5] C. Diwan, S. Srinivasa, G. Suri, S. Agarwal, and P. Ram, "Ai-based learning content generation and learning pathway augmentation to increase learner engagement," *Computers and Education: Artificial Intelligence*, vol. 4, p. 100110, 2023.

[6] M. Forehand, "Bloom's taxonomy," *Emerging perspectives on learning, teaching, and technology*, vol. 41, no. 4, pp. 47–56, 2010.

[7] J. M. Dubas and S. A. Toledo, "Taking higher order thinking seriously: Using marzano's taxonomy in the economics classroom," *International Review of Economics Education*, vol. 21, pp. 12–20, 2016.

[8] B. Ramdurai and P. Adhithya, "The impact, advancements and applications of generative ai," *International Journal of Computer Science and Engineering*, vol. 10, no. 6, pp. 1–8, 2023.

[9] N. Scaria, S. Dharani Chenna, and D. Subramani, "Automated educational question generation at different bloom's skill levels using large language models: Strategies and evaluation," in *International Conference on Artificial Intelligence in Education*. Springer, 2024, pp. 165–179.

[10] S. Narayanan, V. S. Kommuri, N. S. Subramanian, K. Bijlani, and N. C. Nair, "Unsupervised learning of question difficulty levels using assessment responses," in *Computational Science and Its Applications–ICCSA 2017: 17th International Conference, Trieste, Italy, July 3-6, 2017, Proceedings, Part I 17*. Springer, 2017, pp. 543–552.

[11] S. Narayanan, F. M. Saj, K. Bijlani, and S. P. Rajan, "Automatic assessment item bank calibration for learning gap identification," in *2018 International conference on advances in computing, communications and informatics (ICACCI)*. IEEE, 2018, pp. 1429–1435.

[12] H. S. Sætra, "Generative ai: Here to stay, but for good?" *Technology in Society*, vol. 75, p. 102372, 2023.

[13] A. Killawala, I. Khokhlov, and L. Reznik, "Computational intelligence framework for automatic quiz question generation," in *2018 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*. IEEE, 2018, pp. 1–8.

[14] R. Dijkstra, Z. Genç, S. Kayal, J. Kamps *et al.*, "Reading comprehension quiz generation using generative pre-trained transformers." in *iTextbooks@ AIED*, 2022, pp. 4–17.

[15] S. Elkins, E. Kochmar, J. C. Cheung, and I. Serban, "How teachers can use large language models and bloom's taxonomy to create educational quizzes," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 21, 2024, pp. 23 084–23 091.

[16] W. Fu, B. Wei, J. Hu, Z. Cai, and J. Liu, "Qgeval: A benchmark for question generation evaluation," *arXiv preprint arXiv:2406.05707*, 2024.

[17] L. Pan, Y. Xie, Y. Feng, T.-S. Chua, and M.-Y. Kan, "Semantic graphs for generating deep questions," *arXiv preprint arXiv:2004.12704*, 2020.

[18] K. Hwang, S. Challagundla, M. Alomair, L. K. Chen, and F.-S. Choa, "Towards ai-assisted multiple choice question generation and quality evaluation at scale: Aligning with bloom's taxonomy," in *Workshop on Generative AI for Education*, 2023.

[19] P. Laban, C.-S. Wu, L. Murakhovs' ka, W. Liu, and C. Xiong, "Quiz design task: Helping teachers create quizzes with automated question generation," *arXiv preprint arXiv:2205.01730*, 2022.

[20] O. Balamurali, A. A. Sai, M. Karthikeya, and S. Anand, "Sentiment analysis for better user experience in tourism chatbot using lstm and llm," in *2023 9th International Conference on Signal Processing and Communication (ICSC)*. IEEE, 2023, pp. 456–462.

[21] M. Brantner, D. Florescu, D. Graf, D. Kossmann, and T. Kraska, "Building a database on s3," in *Proceedings of the 2008 ACM SIGMOD international conference on Management of data*, 2008, pp. 251–264.

[22] R. Islam and I. Ahmed, "Gemini-the most powerful llm: Myth or truth," in *2024 5th Information Communication Technologies Conference (ICTC)*. IEEE, 2024, pp. 303–308.

[23] H. V. Khurdula, A. Paguthariv u, and J. S. Yoo, "The future of feelings: Leveraging bi-lstm, bert with attention, palm v2 & gemini pro for advanced text-based emotion detection," in *SoutheastCon 2024*. IEEE, 2024, pp. 275–278.

[24] N. Habbat, H. Nouri, H. Anoun, and L. Hassouni, "Sentiment analysis of imbalanced datasets using bert and ensemble stacking for deep learning," *Engineering Applications of Artificial Intelligence*, vol. 126, p. 106999, 2023.

[25] M. Imran and N. Almusharraf, "Google gemini as a next generation ai educational tool: a review of emerging educational technology," *Smart Learning Environments*, vol. 11, no. 1, p. 22, 2024.

[26] G. Feng, B. Zhang, Y. Gu, H. Ye, D. He, and L. Wang, "Towards revealing the mystery behind chain of thought: a theoretical perspective," *Advances in Neural Information Processing Systems*, vol. 36, pp. 70 757–70 798, 2023.

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